

Ahmad AlAmmouri

STAFF RESEARCH ENGINEER · WIRELESS COMMUNICATIONS

Plano, TX, USA

Summary

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Staff Research Engineer with a PhD in Wireless Communications from UT Austin and over a decade of experience designing and optimizing next-generation wireless systems, including mmWave and FR3. Led system-level architecture, MIMO design, and hybrid beamforming development for multiple PoCs, bridging theory and real-world deployment constraints. Co-inventor on 35+ patents and author of 25+ publications. Experienced in driving end-to-end system design, simulation, and cross-functional execution.

Research Experience

Samsung Research America (SRA)

Plano, Texas

STAFF RESEARCH ENGINEER

Mar. 2024 - Now

- Led system-level design of mmWave FWA solutions, developing air-interface algorithms and beamforming strategies that increased supported CPEs per access unit by up to 117% in system-level evaluations.
- Owned system-level design for FR3 PoC, defining MIMO architecture and feature set for MIMO/beamforming, and driving link budget and system-level evaluations under realistic propagation and hardware constraints.
- Led cross-team coordination across three Samsung research hubs, aligning system architecture, feature roadmap, and PoC execution for JPTA, FR3, and FWA initiatives.
- Designed energy-aware multi-device connectivity framework, enabling cooperative communication strategies across devices to improve system efficiency under power constraints.
- Contributed to AI-RAN system design by proposing a delay-aware architecture with hierarchical agents and orchestrators, enabling scalable coordination and latency-aware decision-making in dynamic wireless networks.
- Led investigation of UL-MIMO design for FR3 systems, evaluating codebook-based vs non-codebook-based and coherent vs non-coherent transmission schemes, and proposed multi-TRX FDM approaches to improve system flexibility and performance under practical deployment constraints.

SENIOR RESEARCH ENGINEER

Apr. 2021 - Feb. 2024

- Designed and implemented Joint Phase-Time Array (JPTA) PoC for mmWave/THz systems, enabling frequency-selective analog beams and enabling up to 8x increase in UL scheduling opportunities.
- Developed MU-MIMO strategies for upper-mid band systems, including scheduling, hybrid precoding, and 3GPP-compliant design tradeoffs.
- Designed beam management and codebook strategies for Samsung 5G mmWave base stations under practical deployment constraints.
- Developed and maintained system-level, link-level, and ray-tracing simulators to evaluate beamforming and multi-user performance across realistic channel conditions.

National Institute for Research in Digital Science and Technology (INRIA)

Paris, France

CHATEAUBRIAND RESEARCHER

Jan. 2020 - Jul. 2020

- Data Traffic Dynamics in Cellular Networks.

VISITING RESEARCHER

Apr. 2019 - Jul. 2019

- Data Traffic Dynamics in Cellular Networks.

Samsung Research America (SRA)

Richardson, Texas

RESEARCH INTERN

May 2018 - Aug. 2018

- Beam Codebook Design for 5G Mobile Devices With the Impact of User's Handgrip.

RESEARCH INTERN

May 2017 - Aug. 2017

- Integrated Access and Backhaul for 5G mmWaves Cellular Networks.

Education

The University of Texas at Austin

Austin, Texas

PHD IN ELECTRICAL ENGINEERING

Aug. 2016 - Jan. 2021

- Advisors: Prof. Jeffrey Andrews and Prof. François Baccelli
- Thesis: Analysis of Cellular Networks: Densification and Data Traffic Dynamics
- GPA: 4.00/4.00
- A technical reviewer for 50+ peer-reviewed journal papers and the recipient of three Exemplary Reviewer awards for that.
- A technical program committee (TPC) member for ICC 2018-2024.

- Advisor: Prof. Mohamed-Slim Alouini
- Thesis: Full-Duplex Communications in Large-Scale Cellular Networks
- GPA: 4.00/4.00

The University of Jordan

Amman, Jordan

Aug. 2009 - Jan. 2014

- Graduation Project: Distributed Opportunistic Spectrum Sharing in Cognitive Radio Networks
- GPA: 3.73/4.00, Valedictorian, Excellent with Honor

Technical Skills**Wireless Standards:** 3GPP NR (PHY-layer), ORAN.**Programming:** MATLAB, Python, C++, Git.**Tools:** SLS, LLS, PyTorch, Remcom Wireless InSite, HFSS.**Theoretical Foundations:** Signal Processing, Beamforming, MU-MIMO, Hybrid Precoding, Channel Estimation, Stochastic Geometry, Queuing Theory.**Honors & Awards**

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| 2022-2025 | SRA Inventor Award , High quality patents for X-MIMO and JPTA System Development. |
| 2023 | SRA Individual Award , Projects: JPTA and X-MIMO System Development. |
| 2021, 2023 | SRA Team Award , Project: JPTA System and PoC Development. |
| 2020 | Student Leadership Award , awarded by the WNCG at UT Austin. |
| 2019 | Chateaubriand Fellow , awarded by the Embassy of France in the United States. |
| 2019 | Professional Development Award , awarded by the ECE department at UT Austin. |
| 2017, 2018 | Exemplary Reviewer , IEEE Transactions on Wireless Communications. |
| 2018 | SRA President's Award , Beam Codebooks for mmWaves Mobile Devices. |
| 2017 | Exemplary Reviewer , IEEE Transactions on Communications. |
| 2016 | Best Poster Award , KAUST-NSF Conf. on Electronic Systems for a Sustainable Future. |
| 2014-2016 | King Abdullah University of Science and Technology Fellowship , awarded to only one student from Jordan in the class of 2016. |
| 2014 | University of Jordan Award for Scientific Excellence , top of Electrical Engineering class. |

Selected Publications and Patents**1 [B] Book. 14 [J] Journal paper. 12 [C] Conference paper. 37 [P] Patents.** <Full list available here>**[B1]** J. G. Andrews, A. K. Gupta, **A. AlAmmouri**, and H. S. Dhillon, "An Introduction to Cellular Network Analysis using Stochastic Geometry", Springer, Jun. 2023.**[J4]** **A. AlAmmouri**, J. MO, V. V. Ratnam, B. L. NG, R. W. Heath, J. Lee, J. Zhang, "Extending Uplink Coverage of mmWave and Terahertz Systems Through Joint Phase-Time Arrays", IEEE Access, Aug. 2022.**[J3]** **A. AlAmmouri**, J. Mo, B. L. Ng, J. C. Zhang, and J. G. Andrews, "Hand Grip Impact on 5G mmWave Mobile Devices", IEEE Access, May 2019.**[J2]** **A. AlAmmouri**, J. G. Andrews, and F. Baccelli, "A Unified Asymptotic Analysis of Area Spectral Efficiency in Ultradense Cellular Networks", IEEE Transactions on Information Theory, Jun. 2018.**[J1]** **A. AlAmmouri**, H. ElSawy, O. Amin, and M. -S. Alouini, "In-Band α -Duplex Scheme for Cellular Networks: A Stochastic Geometry Approach", IEEE Transactions on Wireless Communications, Jul. 2016.